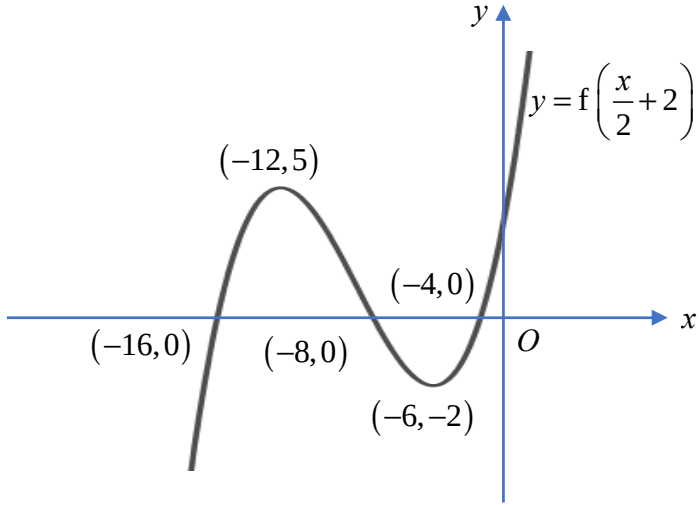
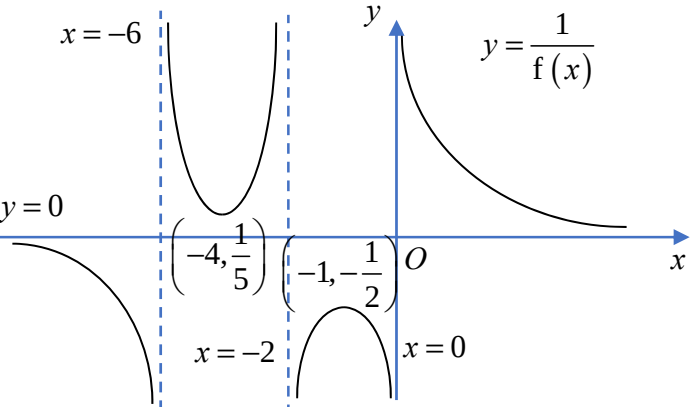
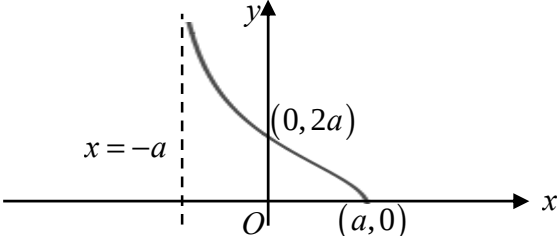
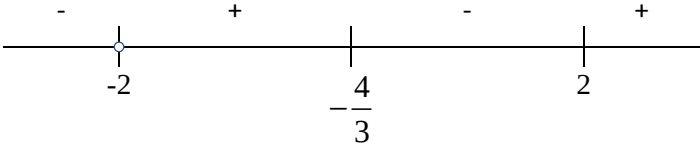
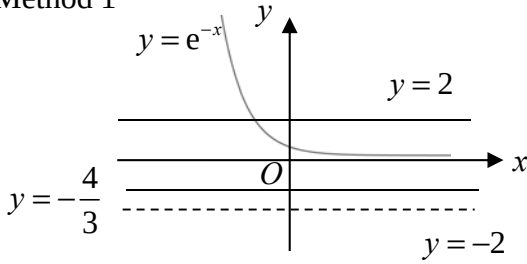


Qn	Solutions
1	Let c, i and l be the amount Abel and Paula spent on a “circuit”, “intervals” and “long run” session in minutes respectively. $\begin{aligned} 23c + 40i + 61l &= 12600 \\ 34c + 67(1.1)i + 75l &= 17600.5 \\ 33c + 53i + 87l &= 17725 \end{aligned}$ By GC, $c = 90$, $i = 65$ and $l = 130$ Abel, Mo and Paula spent 2600 minutes, 4790.5 minutes and 3445 minutes on “Intervals” respectively in the year.
2i	
ii	
3	$\frac{x^2}{64} + \frac{(y-6)^2}{36} = 1$ $x^2 = 64 \left(1 - \frac{(y-6)^2}{36} \right)$

Qn	Solutions
	Volume of bowl = $\pi \int_h^6 \left(1 - \frac{(y-6)^2}{36} \right) dy$ $= 64\pi \left[y - \frac{(y-6)^3}{108} \right]_h^6$ $= 64\pi \left(6 - h + \frac{(h-6)^3}{108} \right)$ $= 384\pi - 64\pi h + \frac{16(h-6)^3}{27}$ $384\pi - 64\pi h + \frac{16(h-6)^3}{27} \geq 790$  From the graph, $0 < x \leq 0.664$ Maximum height of the base is 0.664 units.
4i	 Optional: When $x = 0$, $a \cos 2t = 0 \Rightarrow t = \frac{\pi}{4}$ $y = 2a \tan\left(\frac{\pi}{4}\right) = 2a$ When $y = 0$, $2a \tan t = 0 \Rightarrow t = 0$ $x = a \cos 0 = a$ As $t \rightarrow \frac{\pi}{2}$, $x \rightarrow -a$, $y \rightarrow \infty$ $\therefore x = -a$ is an asymptote.
ii	$x = 3\cos(2t), \quad y = 6\tan t$

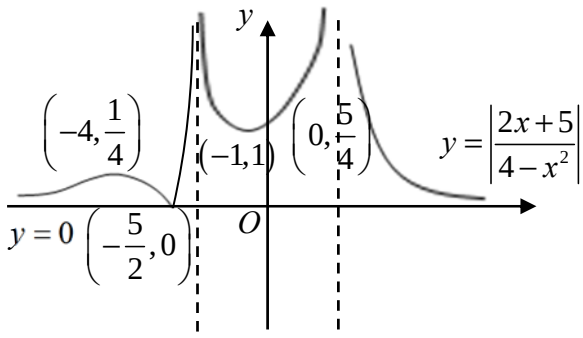
Qn	Solutions
	$\frac{dx}{dt} = -6\sin(2t), \quad \frac{dy}{dt} = 6\sec^2 t$ $\frac{dy}{dx} = -\frac{\sec^2 t}{\sin(2t)}$ When $t = \frac{\pi}{3}$, $x = -\frac{3}{2}$, $y = 6\sqrt{3}$, $\frac{dy}{dx} = -\frac{\frac{2^2}{\sqrt{3}}}{\frac{2}{2}} = -\frac{8}{\sqrt{3}}$ $y - 6\sqrt{3} = -\frac{8}{\sqrt{3}}\left(x + \frac{3}{2}\right)$ $y = -\frac{8}{\sqrt{3}}x - 4\sqrt{3} + 6\sqrt{3}$ $y = -\frac{8}{\sqrt{3}}x + 2\sqrt{3}$ Equation of tangent is $y = -\frac{8}{\sqrt{3}}x + 2\sqrt{3}$
5	$\frac{7x+6}{x+2} \leq 3x-1$ $\frac{(3x-1)(x+2)-7x-6}{x+2} \geq 0$ $\frac{3x^2+5x-2-7x-6}{x+2} \geq 0$ $\frac{3x^2-2x-8}{x+2} \geq 0$ $\frac{(x-2)(3x+4)}{x+2} \geq 0$  $-2 < x \leq -\frac{4}{3} \text{ or } x \geq 2$
	$\frac{7+6e^x}{1+2e^x} \leq 3e^{-x} - 1$ $\frac{7e^{-x}+6}{e^{-x}+2} \leq 3e^{-x} - 1$ Replace x with e^{-x}

Qn	Solutions
	Method 1  $\{x \in \mathbb{R} : x \leq -\ln 2\}$ Method 2 $-2 < e^{-x} \leq -\frac{4}{3} \quad (\text{NAQ } e^{-x} > 0 \text{ for all } x \in \mathbb{R}) \text{ or } e^{-x} \geq 2$ $-x \geq \ln 2$ $x \leq -\ln 2$ $\{x \in \mathbb{R} : x \leq -\ln 2\}$
6a	$\left(\frac{\sqrt{2} + i\sqrt{2}}{1 - i\sqrt{3}} \right)^8 = \left(\frac{2e^{i\frac{\pi}{4}}}{2e^{-i\frac{\pi}{3}}} \right)^8$ $= \left(e^{i\frac{7\pi}{12}} \right)^8$ $= e^{i\frac{14\pi}{3}}$ $= e^{i\frac{2\pi}{3}}$ $= -\frac{1}{2} + \frac{\sqrt{3}}{2}i$
b	$2iz^* - w = 1 + 7i \Rightarrow w = 2iz^* - 1 - 7i$ $z + 2w = 16 - 2i$ $z + 2(2iz^* - 1 - 7i) = 16 - 2i$ Let $z = a + bi$, $a, b \in \mathbb{R}$ $a + bi + 2(2i(a - bi) - 1 - 7i) = 16 - 2i$ $a + bi + 4ai + 4b - 2 - 14i = 16 - 2i$ $a + 4b + i(4a + b) = 18 + 12i$ $a + 4b = 18 \quad \text{--- (1)}$ $4a + b = 12 \quad \text{--- (2)}$

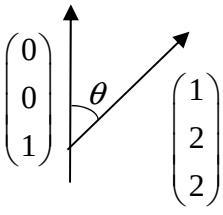
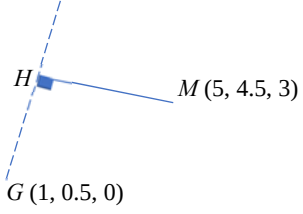
Qn	Solutions
	$(1) \times 4 : 4a + 16b = 72 \Rightarrow (3)$ $(3) - (2) : 15b = 60 \Rightarrow b = 4$ $a = 18 - 4(4) = 2$ $z = 2 + 4i$ $w = 2i(2 - 4i) - 1 - 7i = 7 - 3i$
7	$y = \tan^{-1}(ax + 1)$ $\tan y = ax + 1$ $\sec^2 y \frac{dy}{dx} = a \Rightarrow \frac{dy}{dx} = a \cos^2 y$ $\frac{d^2 y}{dx^2} = -2a \cos y \sin y \frac{dy}{dx} = -a \sin 2y \frac{dy}{dx}$ $\frac{d^3 y}{dx^3} = -2a \cos 2y \left(\frac{dy}{dx} \right)^2 - a \sin 2y \frac{d^2 y}{dx^2}$ When $x = 0, y = \tan^{-1}(1) = \frac{\pi}{4}$ $\frac{dy}{dx} = a \cos^2 \left(\frac{\pi}{4} \right) = \frac{1}{2}a$ $\frac{d^2 y}{dx^2} = -a \sin \left(\frac{\pi}{2} \right) \left(\frac{1}{2}a \right) = -\frac{1}{2}a^2$ $\frac{d^3 y}{dx^3} = 0 - a \sin \left(\frac{\pi}{2} \right) \left(-\frac{a^2}{2} \right) = \frac{a^3}{2}$ $y = \frac{\pi}{4} + \frac{1}{2}ax + \left(-\frac{a^2}{2} \right) \frac{x^2}{2!} + \left(\frac{a^3}{2} \right) \frac{x^3}{3!} + K$ $= \frac{\pi}{4} + \frac{1}{2}ax - \frac{1}{4}a^2x^2 + \frac{1}{12}a^3x^3 + K$ $\sqrt{9-x} = 3 \left(1 - \frac{x}{9} \right)^{\frac{1}{2}}$ $= 3 \left(1 + \frac{1}{2} \left(-\frac{x}{9} \right) + K \right)$ $= 3 - \frac{x}{6} + K$ $b \tan^{-1}(ax + 1) = \frac{\pi b}{4} + \frac{1}{2}abx + K$

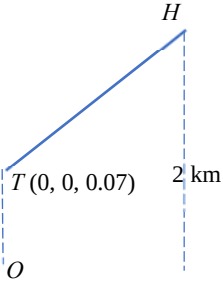
Qn	Solutions
	$\frac{\pi b}{4} = 3 \Rightarrow b = \frac{12}{\pi}$ $\frac{1}{2}a\left(\frac{12}{\pi}\right) = -\frac{1}{6} \Rightarrow a = -\frac{\pi}{36}$
8ai	Equation of l is $r = \begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix} + s \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}, s \in \mathbb{R}$. Equation of p is $r \cdot \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = 6$. Substituting equation of l into equation of p, $\left(\begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix} + s \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} \right) \cdot \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = 6$ $-34 + 20s = 6 \Rightarrow s = 2$ $\underline{\underline{ur}}_{OC} = \begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ $\therefore$ Position vector of C is $3\hat{i} + 2\hat{j} + \hat{k}$ (shown).
ii	Since l is perpendicular to line, $\begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$ is perpendicular to the line. Since p is perpendicular to line, $\begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}$ is perpendicular to the line. Direction vector of line = $\begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} \times \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = \begin{pmatrix} -6 \\ -2 \\ 1 \end{pmatrix}$ Equation of line is $r = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + t \begin{pmatrix} -6 \\ -2 \\ 1 \end{pmatrix}, t \in \mathbb{R}$.
bi	Let θ be the angle between $\underline{a}$ and $\underline{b}$.

Qn	Solutions
	$ \begin{aligned} \left \frac{a \times b}{\% \%} \right ^2 &= \left(\left \frac{a}{\%} \right \left \frac{b}{\%} \right \sin \theta \right)^2 \\ &= \left \frac{a}{\%} \right ^2 \left \frac{b}{\%} \right ^2 \sin^2 \theta \\ &= \left \frac{a}{\%} \right ^2 \left \frac{b}{\%} \right ^2 - \left \frac{a}{\%} \right ^2 \left \frac{b}{\%} \right ^2 \cos^2 \theta \\ &= \left \frac{a}{\%} \right ^2 \left \frac{b}{\%} \right ^2 - \left(\frac{a \cdot b}{\% \%} \right)^2 \text{ (shown)} \end{aligned} $ Alternatively $ \begin{aligned} \left \frac{a}{\%} \right ^2 \left \frac{b}{\%} \right ^2 - \left(\frac{a \cdot b}{\% \%} \right)^2 &= \left \frac{a}{\%} \right ^2 \left \frac{b}{\%} \right ^2 - \left(\left \frac{a}{\%} \right \left \frac{b}{\%} \right \cos \theta \right)^2 \\ &= \left \frac{a}{\%} \right ^2 \left \frac{b}{\%} \right ^2 - \left \frac{a}{\%} \right ^2 \left \frac{b}{\%} \right ^2 \cos^2 \theta \\ &= \left \frac{a}{\%} \right ^2 \left \frac{b}{\%} \right ^2 (1 - \cos^2 \theta) \\ &= \left(\left \frac{a}{\%} \right \left \frac{b}{\%} \right \sin \theta \right)^2 \\ &= \left \frac{a \times b}{\% \%} \right ^2 \text{ (shown)} \end{aligned} $
ii	$ \begin{aligned} \text{Area of } \triangle OAB &= \frac{1}{2} \left \frac{a \times b}{\% \%} \right \\ &= \frac{1}{2} \sqrt{\left \frac{a}{\%} \right ^2 \left \frac{b}{\%} \right ^2 - \left(\frac{a \cdot b}{\% \%} \right)^2} \\ &= \frac{1}{2} \sqrt{1^2 (3)^2 - \left(\frac{1}{3} \right)^2} \\ &= \frac{1}{2} \sqrt{\frac{80}{9}} \\ &= \frac{2\sqrt{5}}{3} \text{ units}^2 \end{aligned} $
iii	$\left \frac{a \cdot b}{\% \%} \right $ is the length of projection of $\frac{b}{\%}$ onto $\frac{a}{\%}$.
9i	
ii	When $y = 2$, $x = 0.82288$ or -1.8229 $ \begin{aligned} \text{Area formed} &= \int_{-1.8229}^{0.82288} \left(2 - \frac{2x+5}{4-x^2} \right) dx \\ &= 1.95 \text{ unit}^2 \end{aligned} $

Qn	Solutions
iii	 From graph, $\left\{ k \in \mathbb{R} : 0 < k < \frac{1}{4} \text{ or } 1 < k < \frac{5}{4} \right\}$.
10ai	Gradient of $PQ = \tan(\pi - \theta) = -\tan \theta$ $y - 5 = -\tan \theta (x - 4)$ $y = (4 - x) \tan \theta + 5$
ii	When $x = 0$, $OQ = 4 \tan \theta + 5$ When $y = 0$, $OP = 4 + 5 \cot \theta$ $OP + OQ = 4 \tan \theta + 5 \cot \theta + 9$ Let $OP + OQ = s$ $\frac{ds}{d\theta} = 4 \sec^2 \theta - 5 \operatorname{cosec}^2 \theta$ At stationary value, $4 \sec^2 \theta - 5 \operatorname{cosec}^2 \theta = 0$ $4 \tan^2 \theta - 5 = 0$ $\tan \theta = \frac{\sqrt{5}}{2} \left(\text{Since } 0 < \theta < \frac{\pi}{2} \right)$ $s = 2\sqrt{5} + 2\sqrt{5} + 5 = 4\sqrt{5} + 5$ $\frac{d^2s}{d\theta^2} = 8 \sec^2 \theta \tan \theta + 10 \operatorname{cosec}^2 \theta \cot \theta$ Since θ is in the first quadrant, $\frac{d^2s}{d\theta^2} > 0$. The stationary value is a minimum value.
b	$\frac{1}{2}ab = 100 \Rightarrow b = \frac{200}{a}$ <u>Method 1</u>

Qn	Solutions
	$c^2 = a^2 + b^2$ $= a^2 + \frac{200^2}{a^2}$ $2c \frac{dc}{dt} = 2a \frac{da}{dt} - \frac{2(200)^2}{a^3} \frac{da}{dt}$ When $a = 20$, $c = \sqrt{20^2 + \frac{200^2}{20^2}} = \sqrt{500}$ $\frac{dc}{dt} = \frac{1}{2\sqrt{500}} \left(2(20)(3) - \frac{2(200)^2}{20^3} (3) \right)$ $= \frac{90}{20\sqrt{5}}$ $= \frac{9}{2\sqrt{5}} \text{ (or 2.01)}$ <u>Method 2</u> $c = \sqrt{a^2 + \frac{200^2}{a^2}}$ $\frac{dc}{da} = \frac{1}{2} \left(a^2 + \frac{200^2}{a^2} \right)^{-\frac{1}{2}} (2a + 200^2 (-2a^{-3}))$ When $a = 20$, $\frac{dc}{dt} = \frac{dc}{da} \cdot \frac{da}{dt}$ $= \frac{1}{2} \left(20^2 + \frac{200^2}{20^2} \right)^{-\frac{1}{2}} \left(40 + 200^2 \left(\frac{-2}{20^3} \right) \right) (3)$ $= 2.01$
11i	When $t = 0$, $\vec{OG} = \begin{pmatrix} 1 \\ 0.5 \\ 0 \end{pmatrix}$ The coordinates of G are (1, 0.5, 0). The flight path is a straight line starting from the point (1, 0.5, 0) moving in the direction of $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$.
ii	Let θ be the angle the flight path makes with the vertical.

Qn	Solutions
	$\cos \theta = \frac{\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}{3} = \frac{2}{3}$ $\theta = 48.190^\circ \text{ (or } 0.84107)$  Required angle = $90^\circ - 48.190^\circ = 41.8^\circ$ (or $\frac{\pi}{2} - 0.84107 = 0.730$) The acute angle the flight path makes with the horizontal is 41.8°.
iii	<u>Method 1</u> $\vec{OH} = \begin{pmatrix} 1 \\ 0.5 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ $\vec{MH} = \begin{pmatrix} -4 \\ -4 \\ -3 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$  $\left(\begin{pmatrix} -4 \\ -4 \\ -3 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right) \cdot \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = 0$ $-18 + 9t = 0 \Rightarrow t = 2$ <u>Method 2</u> $\vec{GM} = \begin{pmatrix} 5 \\ 4.5 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 0.5 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 3 \end{pmatrix}$ Projection of $\vec{GM}$ on $\vec{GH}$ $= \left[\begin{pmatrix} 4 \\ 4 \\ 3 \end{pmatrix} \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right] \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ $= 2 \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ The helicopter will be closest to M 2 min after takeoff.
iv	Equation for flight path of eagle is

Qn	Solutions
	$r = \begin{pmatrix} 2 \\ 4.5 \\ 2 \end{pmatrix} + s \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}, s \in \mathbb{R}$ Suppose the flight paths intersect, $\begin{pmatrix} 1 \\ 0.5 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4.5 \\ 2 \end{pmatrix} + s \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$ $1+t = 2+3s \text{ --- (1)}$ $0.5+2t = 4.5 \text{ --- (2)}$ $2t = 2+s \text{ --- (3)}$ From (2) and (3), $t = 2$ and $s = 2$. Substitute into (1): LHS = -4 and RHS = 1 The flight paths will not intersect.
v	When $z = 2, t = 1$. $\vec{OH} = \begin{pmatrix} 2 \\ 2.5 \\ 2 \end{pmatrix}$ Let top of control tower be T. $\vec{TH} = \begin{pmatrix} 2 \\ 2.5 \\ 1.93 \end{pmatrix}$ $TH = 3.7383$ The controller is able to sight the helicopter. 
12i	When $N \rightarrow K, \frac{dN}{dt} \rightarrow 0$. K is the population size where there is no change in the population. OR K is the population size that the population will eventually approach.

Qn	Solutions
ii	$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right)$ $= rN \left(\frac{K - N}{K} \right)$ $\frac{K}{N(K - N)} \frac{dN}{dt} = r$ $\int \frac{1}{N} + \frac{1}{K - N} dN = rt + C$ $\ln N - \ln K - N = rt + C$ $\frac{N}{ K - N } = Ae^{rt}$ $\frac{N}{K - N} = De^{rt}$ $N = KDe^{rt} - NDe^{rt}$ $N + NDe^{rt} = KDe^{rt}$ $N = \frac{KDe^{rt}}{1 + De^{rt}}$ $= \frac{K}{\frac{1}{D}e^{-rt} + 1}$ Let $B = \frac{1}{D}$ $N = \frac{K}{1 + Be^{-rt}}$ Alternatively $\frac{1}{N(K - N)} \frac{dN}{dt} = \frac{r}{K}$ $\int \frac{1}{\left(\frac{K}{2}\right)^2 - \left(N - \frac{K}{2}\right)^2} dN = \int \frac{r}{K} dt$ $\frac{1}{2\left(\frac{K}{2}\right)} \ln \left \frac{\frac{K}{2} + \left(N - \frac{K}{2}\right)}{\frac{K}{2} - \left(N - \frac{K}{2}\right)} \right = \frac{r}{K} t + C$ $\frac{1}{K} \ln \frac{N}{ K - N } = \frac{r}{K} t + C$

Qn	Solutions
	$\ln \frac{N}{ K - N } = rt + CK$ $\frac{N}{ K - N } = De^{rt}$
iii	When $N = 10000$, $\frac{dN}{dt} = 10000r \left(1 - \frac{10000}{K} \right) = 100$ $100r - \frac{1000000r}{K} = 1$ When $N = 15000$, $\frac{dN}{dt} = 15000r \left(1 - \frac{15000}{K} \right) = 75$ $200r - \frac{3000000r}{K} = 1$ By GC, $r = \frac{1}{50}, \frac{r}{K} = 10^{-6} \Rightarrow K = 20000$ When $t = 0$, $N = \frac{20000}{1+B} = 10000 \Rightarrow B = 1$ $N = \frac{20000}{1 + e^{-0.02t}}$ 